Exercise – *The Chronosphere*

"If you knew Time as well as I do," said the Hatter, "you wouldn't talk about wasting it."

Alice's Adventures in Wonderland *by Lewis Carroll (1832–1898)*

The Hatter, Alice's friend, is in danger! Following the White Queen's advice, Alice is on a quest to find the Chronosphere, a powerful time traveling machine, and use it to travel back in time to save her friend. However, Time, the Chronosphere's keeper, guards it in his castle and forbids her to borrow it. So she has no choice but to steal it. Her plan is to wait for Time to meet with the Red Queen and steal the Chronosphere during their meeting. However, as soon as Alice will activate the Chronosphere, Time will realize immediately, and then the whole army of the Queen will be on Alice's heels. So she needs to find a safe spot where she can activate the Chronosphere and then has enough time to complete the time travel before the army reaches her.

The Queen's army consists of several regiments of armored card soldiers. Each regiment operates in a straight-line formation at all times, where all soldiers move in the same direction and at the same speed. Between regiments, however, direction and speed may vary. The number of card soldiers in a regiment is so large that we may consider each regiment as an infinitely long straight line. Currently, all regiments are stationary and do not move. But as soon as Alice will activate the Chronosphere, they will be alarmed and all start to move towards Alice immediately.

Alice knows where the regiments are currently stationed and what is their speed when they move. So is there a suitable safe spot for her? As a safe spot she may choose any location that can be reached starting from the castle without encountering any of the regiments. If she selects a best possible safe spot, how much time does she have to complete the time travel after activating the Chronosphere and thereby raising the alarm?

Remark. The signed distance $d(q, \ell)$ of a point $q = (q_x, q_y)$ to a line $\ell : ax + by + c = 0$ can be computed by

$$d(q, \ell) = \frac{aq_x + bq_y + c}{\sqrt{a^2 + b^2}}.$$

Input The first line of the input contains the number $t \leqslant 30$ of test cases. Each of the t test cases is described as follows.

- It starts with a line containing three integers c_x c_y n, separated by a space. They denote
  - (c_x, c_y), the castle's location ($|c_x|, |c_y| \leqslant 2^{24}$);
  - n, the number of regiments ($1 \leqslant n \leqslant 2 \cdot 10^3$).
- The following n lines define the regiments. Each regiment is described by four integers a b c v, separated by a space. They denote
  - a, b, c, the coefficients of the equation $ax + by + c = 0$ of the straight line where the regiment is stationed ($|a|, |b|, |c| \leqslant 2^{24}$; you may suppose that $\sqrt{a^2 + b^2} > 0$ is an integer);
  - v, the speed of the regiment ($1 \leqslant v \leqslant 2^8$), meaning that every armored card soldier in this regiment moves a distance of v per second.

You may assume that the n straight lines that represent regiments are pairwise distinct and that the castle is not on any of those lines. You may also assume that any ray emanating from (c_x, c_y) hits at least one regiment.

Output The output for each test case consists of a separate line that contains a single integer that denotes the maximum time in seconds it takes the army to find Alice at her safe spot, rounded down to the nearest integer.

Points There are four groups of test sets, each of which is worth 20 points. For each group there is also a corresponding hidden test set that is worth 5 points. So, you can achieve $4 \cdot 20 + 4 \cdot 5 = 100$ points overall.

1. For the first group of test sets, you may assume that $1 \leqslant n \leqslant 200$, and that every regiment lies on a straight line that is either vertical or horizontal, that is, either $a = 0$ or $b = 0$.
2. For the second group of test sets, you may assume that $1 \leqslant n \leqslant 200$, that for each regiment $ax_s + by_s + c < 0$, and that its speed is one ($v = 1$).
3. For the third group of test sets, you may assume the speed of all regiments is one ($v = 1$).
4. For the fourth group of test sets, there are no additional assumptions.

Corresponding sample test sets are contained in `testi.in/out`, for $i \in \{1, 2, 3, 4\}$.

Sample Input

```
2
1 1 3
3 4 -24 1
0 -1 0 1
-1 0 0 1
0 4 4
4 -3 -2 1
0 2 -4 2
0 1 -6 1
1 0 1 2
```

Sample Output

```
2
1
```